

Determine Metric Weights

User story

As a dashboard user, I want the evaluation criteria weighted transparently so that each state score is balanced, reproducible, and defensible.

Purpose and recommendation

This document completes the four tasks for user story 40: reviewing the core metrics list, defining weighting criteria, assigning preliminary weights, and testing and documenting the final weights. The recommended Fall 2026 starting point is an equal 25 percent weight for each of the four SIECHI operational domains. Equal weighting is the most defensible default until KARA approves a value judgment that one domain should count more than another.

The weights below are a methodology recommendation for sponsor review, not a claim that KARA has already approved them. Outcome measures remain descriptive context. The Fall 2026 scope states that documented structural policy and implementation evidence determines the grade.

Task 1 Review Core Metrics List

Rubric layer	Variable range	Core content	Weighting role
1. CPS Performance	RV-001 to RV-008	Reports, screening, investigations, findings, removals, foster care, outcomes, time and geography	Context and quality checks; do not independently determine a structural grade.
2. CPS Response Pathway	RV-009 to RV-020	Investigation and alternative-response counts and shares, assignment criteria, switching, reopening, findings, recurrence visibility, service participation, refusal consequences	Structural and process evidence feeding the operational domains.
3. Longitudinal History and System Memory	RV-021 to RV-036	Prior reports and placements, repeat reports, prior response paths, record retention, cross-county and cross-state continuity, record access and destruction	Structural evidence about system memory and continuity.
4. SIECHI Structural Assessment	RV-037 to RV-040	Implementation Barriers; Benefit Distribution and Transparency; Family and Child Access; Accountability and Outcomes	Scored domain layer for the preliminary composite.
5. Longitudinal Child and Family Outcomes	RV-041 to RV-050	Education, employment, housing, physical health, mental health, substance use, justice involvement, mortality, parenting, intergenerational CPS contact	Context unless KARA validates a structural interpretation and the pending life-course framework.

Coverage note. The EIM Rubric Variable Map shows that many candidate variables are not yet available in the EIM, and the README states that Layer 5 is waiting on KARA's twelve-measure life-course framework. Missing coverage must therefore remain visible and must not become a zero score.

Scored operational domains

Domain	Working focus
Implementation Barriers	Implementation lag, administrative complexity, fragmented responsibility, workforce or operational constraints, and other barriers between policy and practice.
Benefit Distribution and Transparency	Funding allocation, geographic distribution, eligibility transparency, publication practices, and visibility of who receives benefits.
Family and Child Access	Eligibility, geographic availability, waiting time, documentation burden, response pathways, and practical access to supports and services.
Accountability and Outcomes	Public reporting, reporting frequency, audits, outcome measurement, corrective action, and evidence that the system learns and responds.

Task 2 Define Metric Weighting Criteria

Criterion	Test
Alignment	The indicator measures a condition that the SIECHI operational domain is intended to assess.
Actionability	The state can reasonably change the policy, process, funding, reporting, or implementation condition.
Evidence quality	The measure has direct official evidence, validated evaluation, audit evidence, or sufficiently credible triangulation.
Comparability	Definitions, numerators, denominators, population, period, and geography support the intended use.
Equity relevance	The indicator helps identify differences in access, burden, benefit, safety, or outcomes by race or geography.
Non-duplication	The indicator adds distinct information and does not count the same evidence in multiple domains.
Stability	The indicator can be applied across states and over time without frequent redefinition.
Interpretability	A policymaker or community reader can understand what the indicator means and why it affects the score.

Weighting rules

1. Use the four operational domains as the top-level weighted components. Do not give a raw outcome statistic an independent weight in the structural composite.
2. Within a domain, assign an indicator weight only after its definition, evidence requirement, direction of better performance, and double-counting risk are documented.
3. Do not assign a numeric value to missing evidence. Use Not Scorable and report the score coverage separately.
4. Do not let the amount of available data determine importance. A state should not receive a higher score merely because it publishes more rows, although transparency itself can be evaluated in the appropriate domain.

5. Apply the same approved weights to every state in the same evaluation cycle and record the framework version.
6. Change a weight only through documented methodology review. Recalculate all pilot states after any change.

Task 3 Assign Preliminary Metric Weights

Operational domain	Preliminary weight	Rationale
Implementation Barriers	25%	Policy cannot produce results if implementation is delayed, fragmented, or unsupported.
Benefit Distribution and Transparency	25%	A state must show how resources and benefits are distributed and reported.
Family and Child Access	25%	Eligibility on paper is insufficient if families cannot reach timely, appropriate support.
Accountability and Outcomes	25%	Public reporting, review, corrective action, and measured results are necessary for accountability.
Total	100%	Equal weights avoid an unapproved preference among domains during the pilot.

Scoring formula

Each scored domain uses the project's minus 2 to plus 2 scale. The preliminary composite is calculated as:

$$\text{Composite score} = 0.25(I) + 0.25(B) + 0.25(F) + 0.25(A)$$

I is Implementation Barriers, B is Benefit Distribution and Transparency, F is Family and Child Access, and A is Accountability and Outcomes. The composite remains on the minus 2 to plus 2 scale.

Coverage must be displayed beside the composite. If a required domain is Not Scorable, the public dashboard should not silently redistribute its weight. KARA's methodology lead should approve either a minimum-coverage rule or a no-composite rule for incomplete domain coverage.

Evidence confidence treatment

Confidence	Meaning	Preliminary use
High	Direct official evidence, validated evaluation, formal audit, or clear administrative measurement.	Eligible for scoring after comparability review.
Moderate	Credible evidence requiring interpretation or triangulation.	Eligible with a visible qualification and reviewer note.
Low	Indirect, incomplete, or proxy-dependent evidence.	Display as provisional context; sponsor decides whether it can affect a public score.
Not Scorable	Evidence cannot support a defensible judgment.	No numeric score; do not convert to zero.
Not Yet Reviewed	The record has not completed student review under the rubric.	No numeric score.

Task 4 Test and Document Final Weights

A sensitivity test checks whether reasonable alternative weights would materially change the interpretation. The examples below are hypothetical and are used only to test the method.

Hypothetical profile	I	B	F	A
Profile A Balanced strength	+1	+1	+1	+1
Profile B Strong access weak accountability	0	+1	+2	-1
Profile C Transparent but major access barriers	-1	0	-2	+2

Scenario	I / B / F / A	Profile A	Profile B	Profile C	Order
Equal weights	25 / 25 / 25 / 25	1.00	0.50	-0.25	A > B > C
Access and accountability emphasis	15 / 20 / 30 / 35	1.00	0.45	-0.05	A > B > C
Accountability-heavy	20 / 20 / 20 / 40	1.00	0.20	0.20	A > B = C

Test result. The alternative scenarios preserve the leading profile but materially change the relationship between Profiles B and C. An accountability-heavy formula creates a tie that the equal-weight formula does not. Because the project has not yet approved a reason to privilege one domain, equal weighting minimizes hidden value judgments and is the most transparent pilot choice.

Final preliminary weights for sponsor review

Use 25 percent for each operational domain in the Fall 2026 pilot. Lock the weights before comparing Minnesota, Arizona, and the third state. Publish the weight table, domain scores, evidence-confidence labels, comparability notes, and score coverage with every composite result.

Approval and change control

- Snigdha Sahai confirms that the four operational domains are the correct scored layer and reviews the treatment of Low evidence.
- Allen Huang confirms that the KARA Platform can store domain weight, indicator weight, framework version, score coverage, and Not Scorable status.
- The team applies the same weights to the three pilot states and records any indicator that cannot be scored.
- The team reruns the sensitivity test using real pilot-state domain scores.
- KARA approves or revises the weights before any public grade, award recommendation, or state call-out is published.

Sources

Source note. The KARA/EIM files listed below are the controlling project sources. External government sources were used to clarify public definitions, reporting practices, and inclusive-language principles. State procedures can change, so dates and definitions should be rechecked before publication.

- KARA: State-by-State CPS Policy Comparison and Awards Project Scope and Sprint 0 Direction, September 17, 2026

- KARA EIM: Field Inventory and Data Tracking README, Rubric Variable Map, Metric Comparability, Pick List Reference, and Validation Lists, updated through September 30, 2026
- KARA EIM: Observation Data Schema and Policy Metrics Schema, updated through September 30, 2026